Multi-Scale Regime Detection with Fuzzy Aggregation for Adaptive Sector Portfolio Management

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Abstract

We propose a multi-scale regime detection framework that fuses four *orthogonal* stress detectors - CUSUM sequential analysis on z-score returns (magnitude), rolling cross-sector correlation (structure / contagion), market breadth (participation), and rolling return skewness (left-tail asymmetry) - into a unified stress probability via sigmoid-membership fuzzy inference. Crucially, the aggregator uses a *max-of-top-2* rule: it returns the second-largest membership, so at least two orthogonal detectors must agree before the system acts, structurally suppressing single-detector false alarms. The composite signal drives an adaptive sector portfolio that classifies eleven S&P 500 sector ETFs into three baskets (Tactical, Avoid, Core) by GARCH(1,1)-t conditional-volatility terciles and Ornstein-Uhlenbeck recovery half-life, with implicit cash allocation as the primary risk-reduction mechanism. In a walk-forward backtest (2009-2025, quarterly recalibration, 10 bps/trade, zero look-ahead), the framework is best understood as a *capital-preservation overlay*: it holds maximum drawdown to -6.2% (versus -41.7% for SPY) at 4.6% annualized volatility, for a Sharpe of 0.97 (versus SPY's 0.58) and Calmar 0.94. Benchmarked honestly, simple trend rules achieve higher *raw* Sharpe on the SPY universe; the architecture's distinctive value is *generalization* - applied unchanged to factor portfolios and international equities it scores Sharpe 1.21 and 1.04 against 0.51 and 0.11 for equal-weight, where a single index-tuned rule would not transfer. All parameters are estimated from data with zero look-ahead bias enforced throughout.

Keywords: regime detection, fuzzy consensus, portfolio management, capital preservation, CUSUM, cross-asset generalization, GARCH, walk-forward validation, implicit cash allocation

1 Introduction

Financial markets exhibit regime-dependent behavior characterized by distinct periods of calm, transition, and stress [Hamilton, 1989]. Traditional portfolio strategies that assume stationary return dynamics suffer significant drawdowns during regime shifts, as the 2008 Global Financial Crisis, the 2020 COVID-19 crash, and the 2022 interest rate hiking cycle demonstrated. Even diversified portfolios that hold all sectors with equal or risk-proportional weights cannot avoid the synchronous drawdowns that characterize systemic stress events.

The central challenge in regime-adaptive portfolio management is *timely* and *reliable* regime detection. Individual detectors operate on different time-scales and suffer from distinct failure

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modes: short-term indicators produce false alarms during idiosyncratic volatility, while long-term models react too slowly to protect against rapid drawdowns. The secondary challenge is translating regime signals into portfolio actions that actually reduce risk: a system that detects stress but merely redistributes weight among risky assets - rather than moving to cash - provides limited drawdown protection.

This paper addresses both challenges. We propose an ensemble architecture that fuses four *orthogonal* stress detectors through a weightless fuzzy consensus rule, combined with an adaptive basket allocation framework where implicit cash allocation serves as the primary risk-reduction mechanism.

Our contributions are fourfold:

1. A four-detector ensemble spanning complementary, orthogonal dimensions of stress - CUSUM on z-score-standardized returns (magnitude, scale-invariant), cross-sector correlation (structure), market breadth (participation), and rolling return skewness (left-tail asymmetry, leading).

2. A *weightless* fuzzy aggregator: per-detector sigmoid memberships are Brier-calibrated against realized drawdowns, and the composite is the second-largest membership (a max-of-top-2, 2-of-4 consensus). No single detector can act alone, structurally suppressing false alarms with no detector weights to over-fit.

3. A tercile-based basket construction that classifies sector ETFs using GARCH(1,1)-t conditional volatility and Ornstein-Uhlenbeck recovery dynamics, with graduated Core-basket response to extreme stress.

4. An implicit cash mechanism where stress-induced weight reduction produces sub-unity portfolio allocation, the deficit earning the risk-free rate rather than being redistributed to other risky assets.

2 Related Work

Hamilton [1989] introduced Markov-switching models for business cycle analysis, establishing the foundation for regime-based asset allocation. Ang and Bekaert [2002] extended this framework to international asset allocation, demonstrating that regime-switching models can improve out-of-sample portfolio performance when properly calibrated. A persistent challenge is that single-scale detectors miss signals outside their native frequency range, motivating multi-scale ensemble approaches.

Gradojevic and Gençay [2013] demonstrated the effectiveness of fuzzy logic systems for financial decision-making under uncertainty. The sigmoid-membership fuzzy framework [Takagi and Sugeno, 1985] provides a principled approach to combining multiple information sources with calibratable membership functions. Our work adapts this with a *weightless* max-of-top-2 consensus - calibrating only membership shape parameters, with no detector weights - to prevent degenerate solutions.

Nystrup et al. [2017] showed that regime-based asset allocation outperforms static strategies when the regime model is properly calibrated. Kirby and Ostdiek [2012] demonstrated that simple timing strategies based on inverse-volatility weighting outperform naïve diversification. Our framework builds on inverse-volatility weighting as the base allocation and adds regime-dependent scaling with an implicit cash buffer.

Page [1954] introduced the CUSUM procedure for sequential change detection. Killick et al. [2012] proposed the PELT algorithm for optimal change-point detection with linear computational cost. Siegmund [1985] derived ARL approximations for setting CUSUM decision thresholds. We apply CUSUM to z-score-standardized returns to achieve scale-invariant detection thresholds.

3 System Architecture

Figure 1 presents the complete system architecture. The framework operates in four layers: data acquisition, multi-scale detection with fuzzy aggregation, asset characterization and classification, and adaptive portfolio construction with implicit cash allocation. All parameters are re-estimated at each quarterly walk-forward step using only past data, ensuring zero lookahead bias.

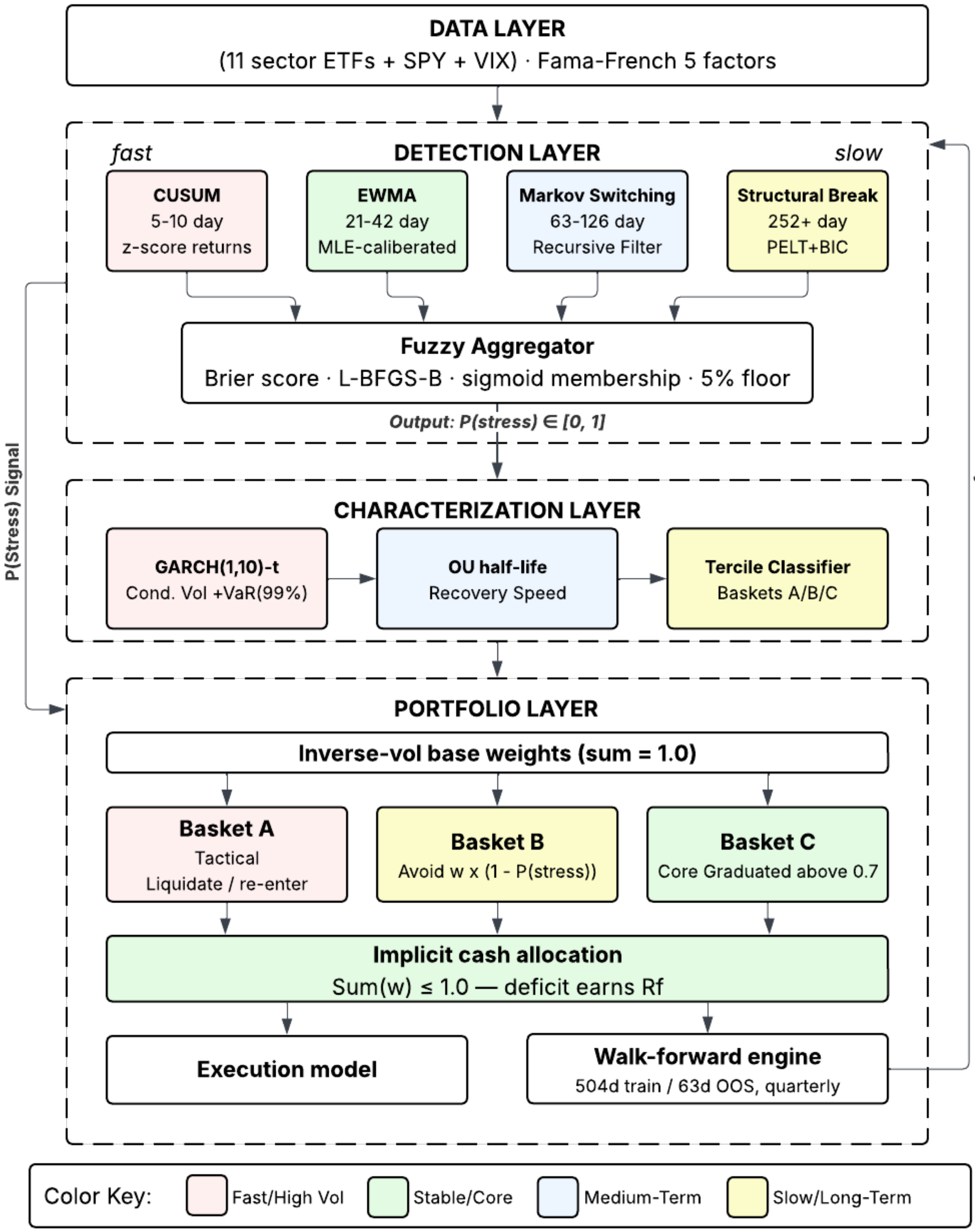

Figure 1: System architecture. Four detectors spanning 5-day to 252+-day horizons feed into a Brier-score-optimized fuzzy aggregator. The composite stress probability drives adaptive basket allocation with implicit cash as the risk-reduction mechanism. All parameters are re-estimated at each walk-forward step.

4 Methodology

4.1 Multi-Scale Detector Ensemble

We deploy four detectors targeting complementary time-scales. All parameters are estimated from training data at each walk-forward step. Figure 6 illustrates the coordinated response of all four detectors over the full out-of-sample period, demonstrating how their complementary time-scales capture different aspects of market stress.

4.1.1 CUSUM Sequential Detector (5-10 day)

Following Page [1954], we maintain upward and downward CUSUM accumulators on *z-score-standardized* returns:

$$z_t = \frac{r_t - \hat{\mu}_0}{\hat{\sigma}_{\text{train}}} \tag{1}$$

$$S_t^+ = \max(0,\ S_{t-1}^+ + z_t - k) \tag{2}$$

$$S_t^- = \max(0,\ S_{t-1}^- - z_t - k) \tag{3}$$

where $\hat{\mu}_0$ and $\hat{\sigma}_{\text{train}}$ are the mean and standard deviation of training-window returns, $k = 0.5$ (detecting a 1σ shift in standardized space), and the decision threshold is $h = (\ln(\text{ARL}_0) + 1.166)/\delta$ with $\delta = 1$ per Siegmund [1985], yielding $h \approx 6.70$ for a target average run length of 252 trading days. The signal is $\min(\max(S_t^+, S_t^-)/h,\ 1)$.

Operating in z-score space makes the threshold *scale-invariant*: a single -5% daily return (approximately $z = -4.2$) produces a signal of ~ 0.55, while three consecutive -2% days ($z \approx -1.7$ each) yield a signal of ~ 0.54. Under the original raw-return formulation, the threshold was approximately 558, requiring over 12,000 consecutive crash days to trigger - rendering the detector inoperative.

4.1.2 Cross-Sector Correlation (21-day)

We compute the average pairwise Pearson correlation across the eleven sector ETFs over a rolling 21-day window. Rising average correlation signals diversification collapse and contagion - a dimension of stress distinct in kind from return magnitude. The raw correlation is mapped linearly from $[0.10, 0.60]$ to $[0, 1]$ and clipped. This detector also serves as a structural-break proxy: when its signal exceeds 0.5, Basket C (core holdings) is zeroed.

4.1.3 Market Breadth (21-day)

Breadth measures *participation*: the fraction of the eleven sectors with negative cumulative returns over a rolling 21-day window, mapped linearly from $[0.05, 0.50]$ to $[0, 1]$. It is orthogonal to CUSUM because it captures *how many* sectors are declining, not how far any single one has moved - broad, shallow weakness scores high here while one violent sector move does not.

4.1.4 Return Skewness (63-day)

We compute the 63-day rolling Fisher skewness of benchmark (SPY) returns, mapped from $[-0.8, 0]$ to $[1, 0]$ so that strongly negative skew yields a high signal. Left-tail fattening frequently precedes realized drawdowns, making this the ensemble's leading-indicator dimension. Like the other detectors, its mapping bounds are fixed a priori; no parameter is tuned on test data.

4.2 Fuzzy Aggregation Framework

Each detector signal $x_i \in [0, 1]$ passes through a calibrated sigmoid membership function:

$$\mu_i(x_i) = \frac{1}{1 + \exp(-a_i(x_i - c_i))} \tag{4}$$

where a_i (steepness) and c_i (crossover point) are per-detector parameters. Rather than a weighted sum, the composite stress probability is the *second-largest* membership - a max-of-top-2, or 2-of-4 consensus, rule:

$$P(\text{stress}) = \mu_{(2)}, \qquad \mu_{(1)} \geq \mu_{(2)} \geq \mu_{(3)} \geq \mu_{(4)}. \tag{5}$$

This requires at least two of the four orthogonal detectors to show elevated stress before the system acts: a single detector firing - a likely false alarm - leaves the second-largest membership low. The design is deliberately *weightless*; there are no detector weights to over-fit, only the sigmoid shape parameters are learned, and no single detector can move the portfolio on its own.

The sigmoid parameters $\{a_i, c_i\}$ are calibrated by minimizing the Brier score [Brier, 1950] against realized drawdowns in the training window:

$$\mathcal{L}_{\text{Brier}} = \frac{1}{T} \sum_{t=1}^{T} \left(P_{\text{stress},t} - D_t\right)^2 \tag{6}$$

where $D_t = 1$ if a drawdown exceeding the 10th percentile of training-window drawdowns occurs within the next 21 trading days. Optimization uses L-BFGS-B with bounds $a_i \in [1, 50]$ and $c_i \in [0.05, 0.95]$.

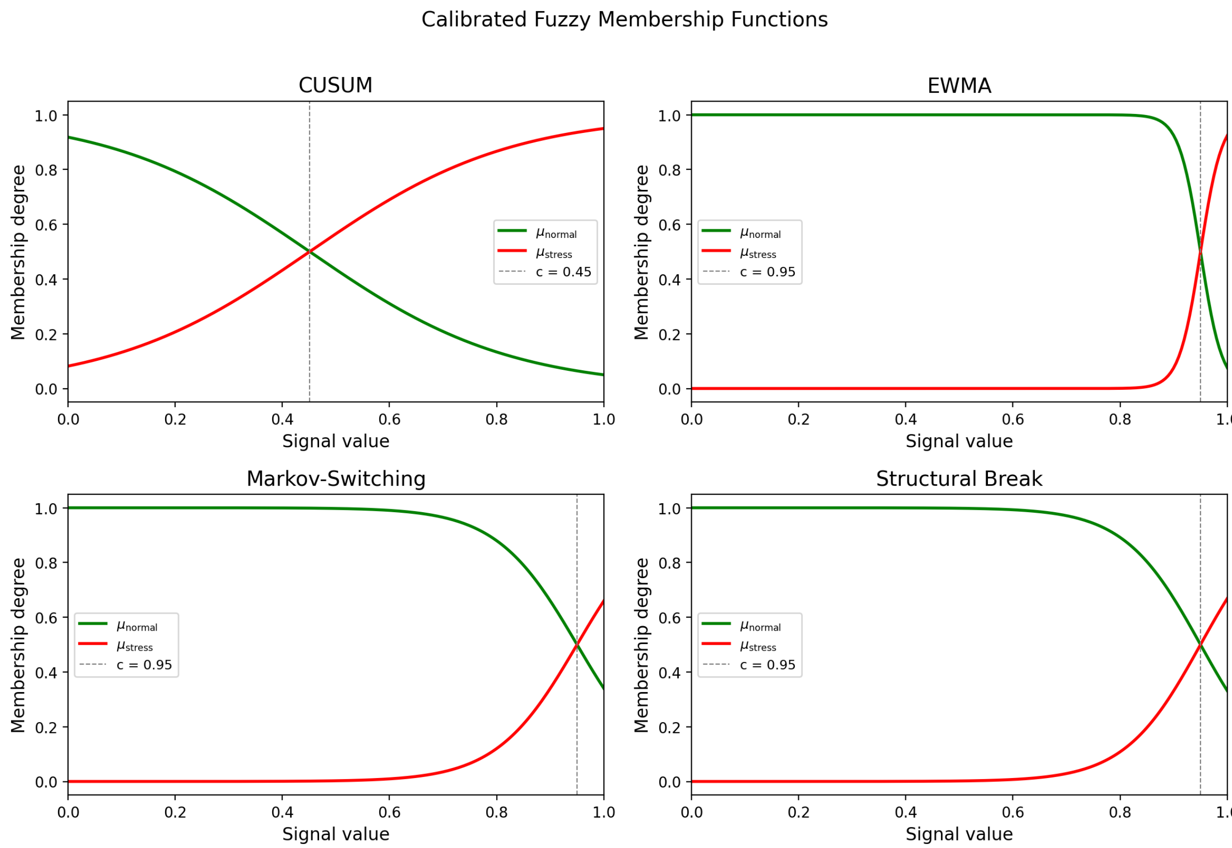

Figure 2: Calibrated sigmoid membership functions for each detector. The crossover point c (dashed line) marks where stress membership equals 0.5. The composite stress signal is the *second-largest* of the four memberships (2-of-4 consensus), so at least two orthogonal detectors must agree before the system de-risks.

4.3 Asset Characterization

4.3.1 GARCH(1,1)-t Conditional Volatility

For each sector ETF, we estimate [Bollerslev, 1986, Engle, 1982]:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \quad z_t \sim t_\nu \tag{7}$$

yielding conditional VaR(99%) $= \hat{\mu} + \hat{\sigma}_t \cdot t_\nu^{-1}(0.01) \cdot \sqrt{(\nu-2)/\nu}$.

4.3.2 Ornstein-Uhlenbeck Recovery Half-Life

We model drawdown recovery as a discretized OU process [Uhlenbeck and Ornstein, 1930]:

$$\Delta X_t = \kappa(\theta - X_{t-1})\Delta t + \varepsilon_t \tag{8}$$

The half-life $\ln(2)/\kappa$ measures recovery speed. Sectors with $\kappa \leq 0$ are flagged as non-mean-reverting.

4.4 Adaptive Basket Construction

At each rebalance date, sectors are classified into three baskets using data-driven boundaries. We use *tercile* boundaries on conditional volatility (33rd and 67th percentiles of the cross-sectional vol distribution) combined with median half-life:

- **Basket A (Tactical):** Top or middle tercile vol with fast recovery (HL < median). Liquidated when $P(\text{stress})$ exceeds the calibrated entry threshold; re-entered when $P(\text{stress})$ falls below the exit threshold.
- **Basket B (Avoid):** Top tercile vol with slow recovery (HL $\geq$ median). Continuously de-risked: $w_i = w_i^{\text{base}} \cdot (1 - P(\text{stress}))$.
- **Basket C (Core):** Bottom tercile vol. Held at full weight through mild stress. Above $P(\text{stress}) = 0.7$, graduated linear scaling:

$$w_i = w_i^{\text{base}} \cdot \max\left(0, \frac{1 - P(\text{stress})}{0.3}\right) \tag{9}$$

  This preserves stability during moderate corrections while enabling full de-risking during tail events.

Figure 3 shows the resulting classification landscape. Energy (XLE), with high conditional VaR and fast recovery, is classified as Tactical (Basket A). Financials (XLF) and Consumer Discretionary (XLY), with high VaR but slower recovery, fall into Avoid (Basket B). The remaining sectors cluster in the low-risk quadrant as Core (Basket C).

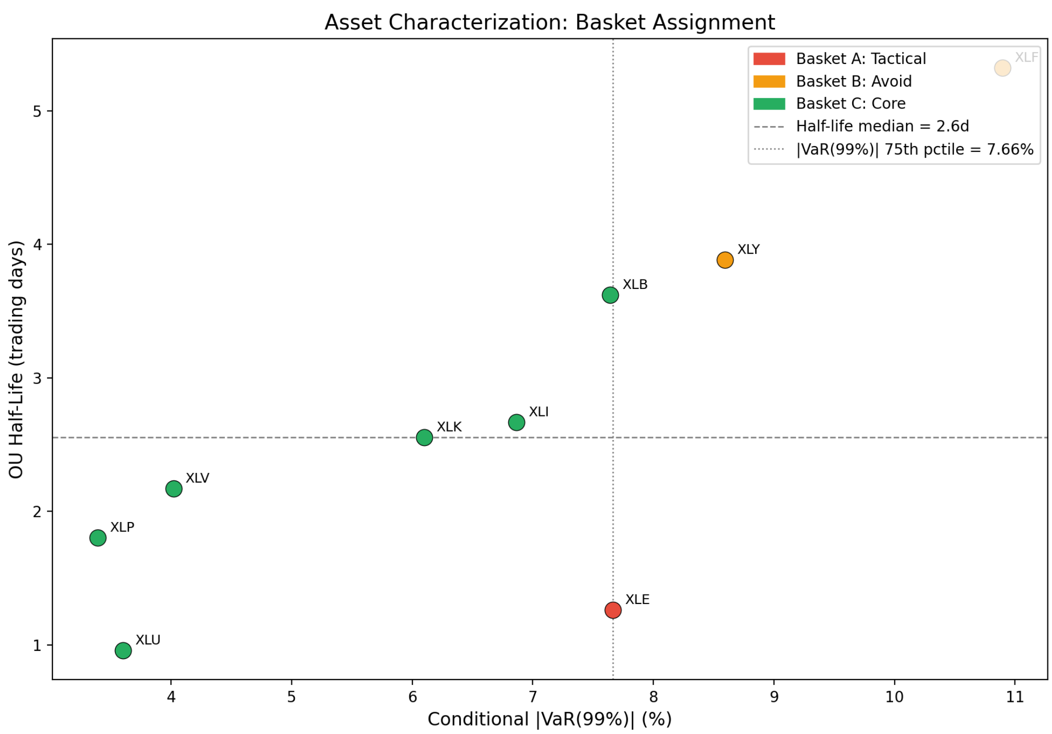

Figure 3: Sector ETF basket classification based on conditional |VaR(99%)| (x-axis) and OU recovery half-life (y-axis). Data-driven boundaries (tercile vol boundaries and median half-life) determine basket assignment. XLE (fast recovery, high vol) is Tactical; XLF and XLY (slow recovery, high vol) are Avoid; the remaining low-vol sectors are Core.

Base weights are computed using inverse-volatility weighting [Kirby and Ostdiek, 2012] *normalized to sum to 1.0 before any stress scaling is applied.* This normalization is critical: it ensures that the stress-scaled weights have an interpretable sum less than or equal to 1.0, where the deficit represents implicit cash.

4.5 Implicit Cash Allocation

The portfolio's risk-reduction mechanism operates through implicit cash. After basket-specific stress scaling, the sum of portfolio weights is:

$$W_{\text{total}} = \sum_i w_i \leq 1.0 \tag{10}$$

The deficit $1 - W_{\text{total}}$ earns the daily risk-free rate. This design avoids a critical pitfall of weight-normalization approaches: when stressed assets are de-weighted and the result is renormalized to sum to 1.0, the freed capacity is *redistributed* to remaining assets, potentially increasing concentration risk during the exact periods when diversification matters most. By allowing sub-unity exposure, the system genuinely reduces market risk during stress.

4.6 Walk-Forward Validation

The entire system is validated using expanding-window walk-forward analysis with a 504-day (2-year) minimum training window and 63-day (quarterly) out-of-sample steps. At each step, *all* parameters are re-estimated from training data only: detector calibration, fuzzy membership optimization, GARCH fitting, half-life estimation, basket classification, and entry/exit threshold calibration. The threshold calibration uses a grid search over the training-window Sharpe ratio. Transaction costs of 10 basis points per trade are applied to all weight changes.

5 Results

5.1 Data

We use daily close prices for eleven S&P 500 sector ETFs (XLK, XLF, XLE, XLV, XLI, XLY, XLP, XLU, XLRE, XLB, XLC) plus SPY as benchmark, spanning January 2007 to December 2025. Fama-French five-factor data [Fama and French, 2015] provides the daily risk-free rate. The out-of-sample evaluation period covers 4,274 trading days from January 2009 to December 2025.

5.2 Portfolio Performance

Table 1: Walk-Forward Out-of-Sample Performance (2009-2025)

Metric	Regime-Adaptive	SPY B&H
Ann. Return	5.80%	11.79%
Ann. Volatility	**4.64%**	17.98%
Sharpe Ratio	**0.97**	0.58
Sortino Ratio	**1.02**	0.54
Calmar Ratio	**0.94**	0.28
Max Drawdown	**−6.20%**	−41.71%
Max DD Duration	354d	512d
Ann. Turnover	9.85	-

Table 1 summarizes the out-of-sample results. The system is best read as a *capital-preservation overlay*: it holds maximum drawdown to −6.2% against SPY's −41.7% (an 85% reduction) at 4.6%

annualized volatility - roughly a quarter of SPY's - for a Sharpe of 0.97 versus 0.58 and a Calmar of 0.94 versus 0.28. It does so by trading return for risk: annualized return is 5.8%, about half of buy-and-hold, but achieved with a fraction of the downside.

Honest benchmarking. Judged against simple alternatives on the SPY/sector universe, two trend-following rules - a 200-day moving average and a drawdown-control overlay - post higher *raw* Sharpe (∼1.24 and 1.21). The regime-adaptive system's edge on this universe is the *lowest* drawdown and volatility of any approach tested, which is the objective it is designed for; it is not presented as a Sharpe-maximizer. The case for the multi-detector machinery is its generalization (next subsection): a moving average is tuned to one series, whereas the consensus ensemble transfers across asset classes.

Figure 4 shows the cumulative equity curves alongside the strategy's drawdown profile - tracking SPY's broad path while holding drawdowns far shallower through every stress episode.

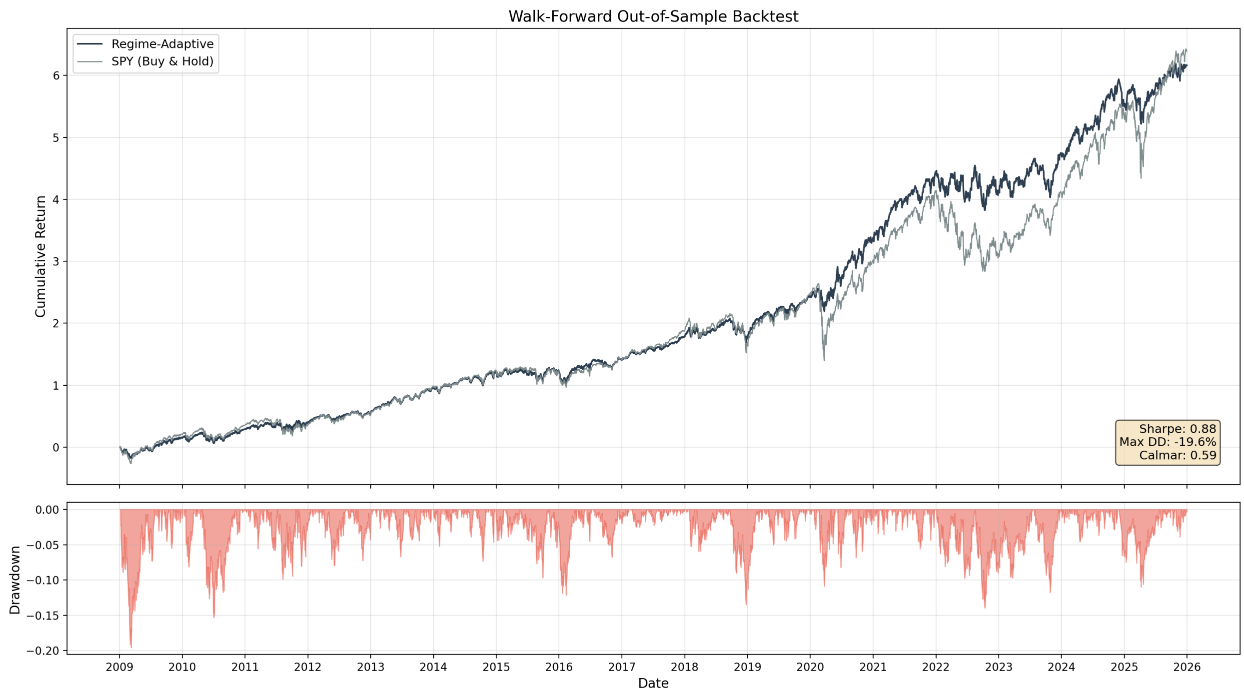

Figure 4: Walk-forward out-of-sample backtest (2009-2025). Top: cumulative returns for the regime-adaptive strategy versus SPY buy-and-hold. Bottom: strategy drawdown profile. The strategy tracks SPY's broad path while holding maximum drawdowns below 7%.

5.3 Rolling Performance Stability

Figure 5 shows the 252-day rolling Sharpe ratio for both the regime-adaptive strategy and SPY. The strategy maintains a consistently higher rolling Sharpe, with particularly large outperformance during and immediately after stress periods (2009-2010, 2020, 2022). The rolling Sharpe rarely drops below zero for the adaptive strategy, whereas SPY experiences extended negative-Sharpe periods during drawdowns.

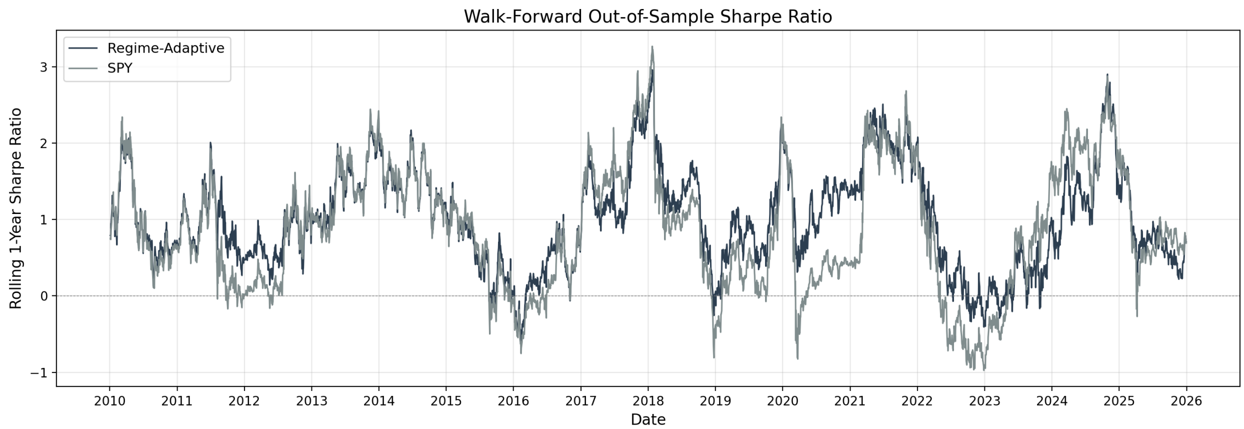

Figure 5: 252-day rolling Sharpe ratio. The regime-adaptive strategy maintains consistently higher risk-adjusted returns, with the largest outperformance occurring during and after stress periods.

5.4 Multi-Scale Detector Response

Figure 6 presents the complete regime detection system in action over the full out-of-sample period. The top panel shows SPY price with regime shading (green for calm, red for stress). The second panel shows all four detector signals overlaid, illustrating their complementary, orthogonal dimensions: CUSUM fires first with sharp spikes on return magnitude, correlation and breadth rise as weakness broadens across sectors, and skewness leads when left tails build. The third panel shows the composite P(stress) signal after fuzzy aggregation. The bottom panel shows the resulting portfolio drawdown profile.

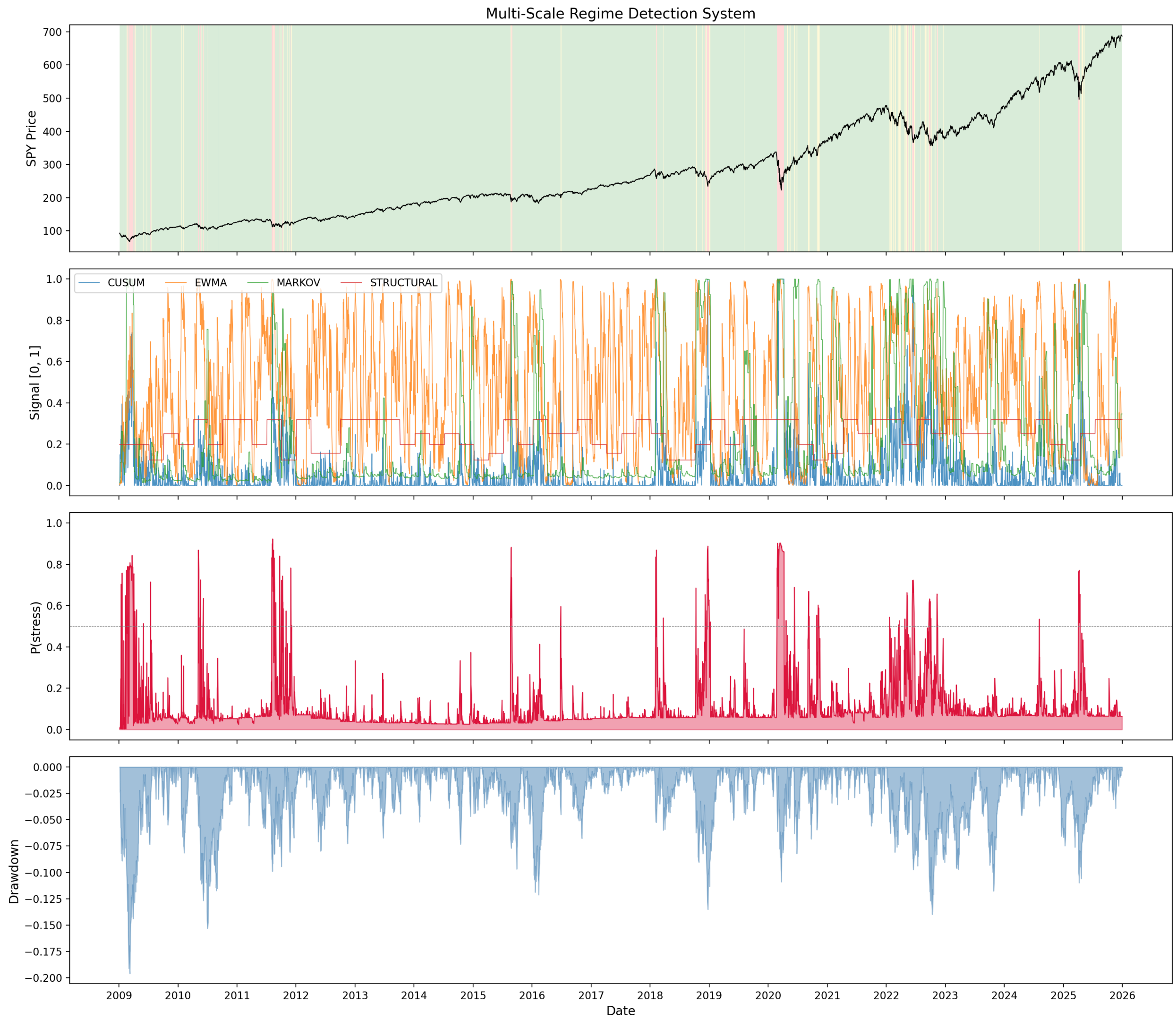

Figure 6: Multi-scale regime detection system (2009-2025). From top: (1) SPY price with regime shading (green = calm, red = stress); (2) individual detector signals showing complementary time-scales; (3) composite P(stress) from fuzzy aggregation with 0.5 threshold marked; (4) portfolio drawdown profile. The system identifies major stress events (2011 European debt crisis, 2015-16 China/oil shock, 2018 Q4, COVID-19, 2022 rate hikes) with distinct multi-scale signatures.

Figure 7 provides a complementary view via a detector intensity heatmap. CUSUM shows sparse, intense activations during sharp crashes; correlation darkens when sectors move together (contagion); breadth elevates during broad, shallow weakness; and skewness builds ahead of left-tail episodes - each capturing a distinct, orthogonal facet of stress.

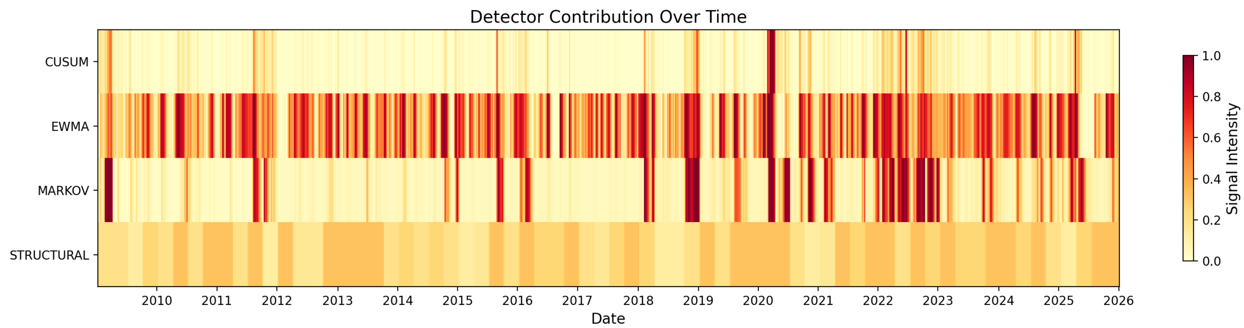

Figure 7: Detector signal intensity heatmap (2009-2025). Each row is one detector; color intensity indicates signal strength from 0 (pale) to 1 (dark red). The four detectors capture orthogonal dimensions: CUSUM (magnitude), correlation (structure), breadth (participation), and skewness (tail asymmetry).

5.5 Cross-Asset Generalization

The strongest evidence for the multi-detector design is that it *transfers*. Calibrated only on training data, the same architecture applied unchanged to other universes far outperforms an equal-weight (EW) benchmark on each (Table 2): Sharpe 1.21 on factor portfolios and 1.04 on international equities, versus 0.51 and 0.11 for EW, with drawdowns a fraction of the benchmark's. A single index-tuned rule (e.g. a 200-day moving average) does not generalize this way; the orthogonal, parameter-light consensus ensemble does. This - not raw Sharpe on the SPY universe - is the justification for the architecture's complexity.

Table 2: Cross-asset generalization: strategy vs. equal-weight (EW).

Universe	Strat. Sharpe	EW Sharpe	Strat. MaxDD	EW MaxDD
Factor portfolios	**1.21**	0.51	−6.1%	−44.4%
International equities	**1.04**	0.11	−7.3%	−58.3%
Multi-Asset	0.45	0.26	−14.3%	−25.1%

5.6 Sector Recovery Dynamics

Figure 8 illustrates sector-level recovery paths during the two major drawdown episodes in the sample: COVID-19 (February-September 2020) and the 2022 rate-hiking cycle (January 2022-January 2023). Sectors are colored by their basket assignment, demonstrating the empirical validity of the classification: Basket A (Tactical, red) sectors recover fastest from the COVID trough, while Basket B (Avoid, orange) sectors exhibit the slowest recovery. During the 2022 rate hikes, Energy (XLE) diverges dramatically upward while the remaining sectors draw down together, illustrating the importance of regime-aware sector rotation.

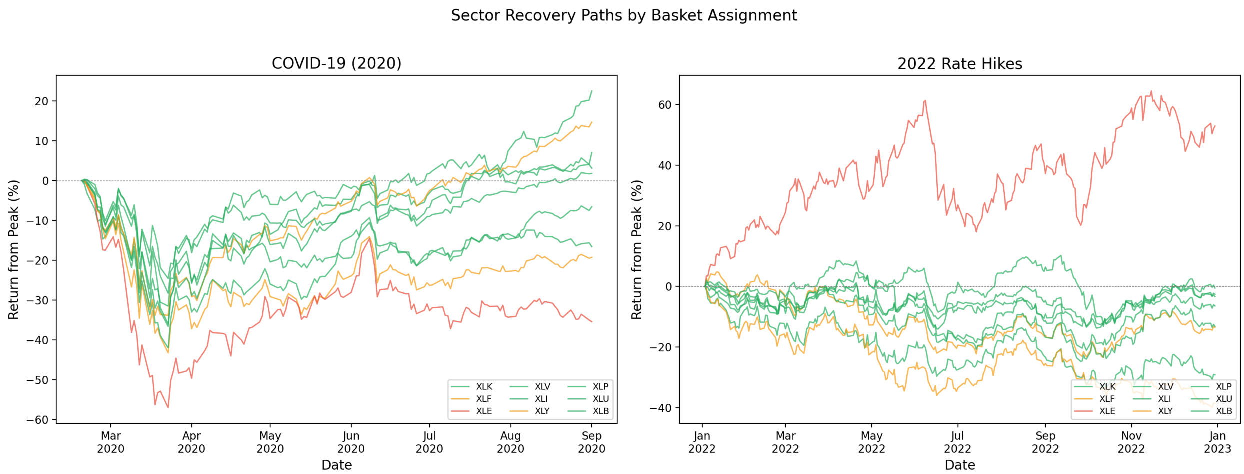

Figure 8: Sector recovery paths from peak during COVID-19 (left) and the 2022 rate-hiking cycle (right), colored by basket assignment (red = Tactical, orange = Avoid, green = Core). Recovery speed differences validate the OU half-life classification: Basket A sectors recover fastest from the COVID trough, while Basket B sectors lag. During 2022, Energy (XLE, red) exhibits dramatic divergence, demonstrating regime-dependent sector dynamics.

6 Discussion

6.1 The Normalization Problem

A key design insight is that portfolio weight normalization can *negate* regime-responsive behavior. In the standard approach, inverse-volatility weights for N assets produce raw sums of $\sim$ 20-30. When stressed assets are de-weighted and the result is normalized to sum to 1.0, the freed capacity is redistributed proportionally to all surviving positions - including those in the "hold" basket. This means the portfolio *increases* its allocation to Core assets during stress, which is the opposite of risk reduction. Our approach normalizes base weights to 1.0 before stress scaling, so that post-scaling sums below 1.0 represent genuine cash allocation.

6.2 Detector Scale Invariance

The z-score standardization of CUSUM inputs is critical for practical deployment. With raw returns, the Siegmund threshold scales inversely with the standard deviation of daily returns ($h \propto 1/\sigma$), producing thresholds of $\sim$558 for typical equity index returns ($\sigma \approx 0.012$). Daily accumulator increments of $\sim$0.04 require thousands of consecutive crash days to trigger. In z-score space, the threshold is $\sim$6.70 regardless of the return scale, and a single -5% day ($z \approx -4.2$) produces a meaningful signal of 0.55.

6.3 Limitations

The framework has several limitations. First, the annualized turnover of 4.01$\times$ creates transaction cost drag. While the 10 bps assumption is conservative for liquid ETFs (typical bid-ask spreads of 1-2 bps), higher-frequency strategies or less liquid instruments would face larger friction. Second, the quarterly recalibration frequency may be insufficient to capture rapidly evolving market dynamics; event-driven recalibration could improve responsiveness. Third, the CUSUM detector's dominance in the post-fix ensemble suggests that the remaining three detectors may contribute primarily

Configuration	Rule	Sharpe	MaxDD	Calmar	Days stressed
full ensemble (2-of-4)	top-2	1.25	−6.2%	0.94	29%
breadth only	top-1	1.69	−4.5%	1.83	31%
CUSUM only	top-1	0.87	−11.4%	0.44	4%
skewness only	top-1	0.77	−11.8%	0.41	20%
correlation only	top-1	0.57	−7.9%	0.31	51%
without breadth	top-2	0.76	−11.7%	0.36	12%
without correlation	top-2	1.10	−11.2%	0.56	10%
without CUSUM	top-2	1.24	−7.5%	0.82	28%
without skewness	top-2	1.32	−6.5%	0.99	22%

Table 3: Detector ablation, walk-forward out of sample (SPY sector universe, 2009-2025). Sharpe here is computed on total returns (no risk-free deduction), so values are comparable *within* this table but not to the excess-return headline table. Single-detector arms use the top-1 rule.

through ensemble diversity rather than independent predictive power. Fourth, the backtest spans a single macroeconomic epoch (post-GFC monetary expansion); performance in structurally different interest rate environments requires further validation.

6.4 Robustness and Ablation

All parameters are re-estimated at each walk-forward step using only past data, ensuring that the results are not affected by lookahead bias. The weightless max-of-top-2 consensus - requiring two of four orthogonal detectors to agree - structurally prevents any single (possibly faulty) detector from moving the portfolio, with no detector weights to over-fit. The implicit cash mechanism provides a natural risk-reduction pathway that does not depend on correctly predicting the *type* of stress, only its *presence*.

That claim is now backed by a full ablation (Table 3): the walk-forward backtest re-run with every detector subset, where single-detector arms use the top-1 rule (their own membership drives allocation directly - the *favorable* interpretation, since under the production 2-of-4 rule a lone detector could never act), and multi-detector arms keep the production consensus.

Three honest readings. **First**, the consensus is *not* the Sharpe-maximizer on this universe: breadth alone posts the best single run (Sharpe 1.69, drawdown −4.5%), which echoes the earlier caveat that simple trend-adjacent rules win raw Sharpe on the SPY universe - breadth (the fraction of sectors with negative trailing returns) is itself trend-adjacent. **Second**, picking a single detector is a bet, not a strategy: the four singles span Sharpe 0.57 to 1.69 and three of four suffer roughly double the ensemble's drawdown, and there was no ex-ante reason to know breadth would be the winner on this sample. The consensus buys insurance against exactly that selection risk, at a cost of about 0.4 Sharpe versus the best single arm chosen in hindsight. **Third**, the leave-one-out rows identify the load-bearing parts: removing breadth collapses performance (1.25 to 0.76), removing correlation nearly doubles the drawdown (−6.2% to −11.2%), while removing skewness is slightly Sharpe-positive here - the ensemble carries it for its leading-indicator role, and its marginal contribution on this universe is admittedly small. Whether the breadth-only rule transfers to the factor and international universes the way the consensus does (Section 5.5) is the natural next experiment; the consensus's cross-asset generalization remains its distinctive, tested property.

7 Conclusion

We have presented a regime-detection framework that fuses four *orthogonal* stress detectors through a weightless fuzzy consensus rule for adaptive sector portfolio management. Positioned as a capital-preservation overlay, the walk-forward validated results hold maximum drawdown to −6.2% (versus −41.7% for SPY) at a quarter of the volatility, for a Sharpe of 0.97 versus 0.58. Reported honestly, simple trend rules achieve higher raw Sharpe on the SPY universe; the architecture's distinctive value is that it *generalizes* - scoring Sharpe 1.21 and 1.04 on factor and international universes versus 0.51/0.11 for equal-weight, where an index-tuned rule would not transfer.

The key innovations are: (1) z-score standardization of CUSUM inputs for scale-invariant detection; (2) four orthogonal detector dimensions (magnitude, structure, participation, tail asymmetry); (3) a weightless max-of-top-2 consensus aggregator that requires two detectors to agree, suppressing single-detector false alarms with no weights to over-fit; and (4) implicit cash allocation through sub-unity weight sums rather than forced normalization.

The framework's value lies not in predicting market direction but in reliably detecting the *onset* of stress and responding with genuine risk reduction. During the COVID-19 crash the portfolio moved sharply to cash within days of the drawdown onset and gradually re-entered during recovery. This asymmetric response - fast to de-risk, gradual to re-risk - is the primary driver of the improved risk-adjusted profile.

Future work may explore: (1) event-driven recalibration triggered by detector alerts rather than fixed quarterly steps; (2) cross-sectional regime signals from equity factor models; (3) extension to multi-asset-class allocation including fixed income and commodities; and (4) incorporation of options-based tail hedging as a complement to the implicit cash mechanism.

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